2023 Winter Qual

III.

$$\frac{\bar{y}_1 - \bar{y}_2 - (\mu_1 - \mu_2)}{\sigma\sqrt{\tilde{n}^{-1}}} \sim N(0,1)$$

$$\frac{\hat{n} S_p^2}{\sigma^2} = \frac{(n_1-1)S_1^2}{\sigma^2} + \frac{(n_2-1)S_2^2}{\sigma^2} = \chi^2(n_1-1) + \chi^2(n_2-1)$$
$$\sim \chi^2(n_1+n_2-2)$$

$$\Rightarrow \frac{\hat{n} S_p^2}{\sigma^2}/\hat{n} \sim \chi^2(n_1+n_2-2)/\hat{n}$$

$$\bar{y}_1, \bar{y}_2 \perp\!\!\!\perp S_1^2, S_2^2 \Longrightarrow \bar{y}_1 - \bar{y}_2 \perp\!\!\!\perp S_p^2$$

$$t = \frac{\dfrac{\bar{y}_1 - \bar{y}_2 - (\mu_1 - \mu_2)}{\sigma\sqrt{\tilde{n}^{-1}}}}{\sqrt{\dfrac{\hat{n} S_p^2}{\sigma^2}/\hat{n}}} = \frac{Z}{\sqrt{\chi^2(\hat{n})/\hat{n}}}$$
$$\sim t(n_1+n_2-2)$$

$$t^2 \sim F(1, \hat{n})$$

$\blacksquare$

$\boxed{3}$.

(a) $\hat{\beta} = (X^TX)^{-1}X^TY$, $X = (\mathbf{1}\ \ X_1)$

$$= \begin{pmatrix} \mathbf{1}^T\mathbf{1} & \mathbf{1}^TX_1 \\ X_1^T\mathbf{1} & X_1^TX_1 \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{1}^T \\ X_1^T \end{pmatrix} y$$

$$= \begin{pmatrix} * & * \\ (3) & (4) \end{pmatrix} \begin{pmatrix} \mathbf{1}^T \\ X_1 \end{pmatrix} y = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix}$$

$$(3) = -[X_1^T(I-H_1)X_1]^{-1} X_1^T\mathbf{1}(\mathbf{1}^T\mathbf{1})^{-1}$$

$$(4) = [X_1^T(I-H_1)X_1]^{-1}$$

$$(3)\mathbf{1}^T + (4)X_1 = [X_1^T(I-H_1)X_1]^{-1}X_1^T(I-H_1) = (X_c^TX_c)^{-1}X_c$$

$$\Rightarrow \hat{\beta}_1 = (X_c^TX_c)^{-1}X_c^TY = \hat{\beta}_c$$

$\square$

(b) $$R^2 = \frac{SSR}{SST} = \frac{Y^T(H_x - H_1)Y}{Y^T(I-H_1)Y}$$

$$\sum (y_i - \bar{y})^2 = ((I-H_1)Y)^T((I-H_1)Y) = Y^T(I-H_1)Y$$

$$\hat{\beta}_c^T X_c^T X_c \hat{\beta}_c = Y^T H_{x_c} Y = Y^T(H_x - H_1)Y$$

$$H_x = X(X^TX)^{-1}X^T = (1\ \ X_c)\begin{pmatrix} \mathbf{1}^T\mathbf{1} & 0 \\ 0 & X_c^TX_c \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{1}^T \\ X_c^T \end{pmatrix}$$

$$= H_1 + H_{x_c} \quad \rightsquigarrow \quad H_{x_c} = H_x - H_1$$

So, $$R^2 = \frac{\hat{\beta}_c^T X_c^T X_c \hat{\beta}_c}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

$$n\bar{y} = \mathbf{1}^TY \Rightarrow n\bar{y}^2 = (\mathbf{1}^TY)^T(\mathbf{1}^T\mathbf{1})^{-1}(\mathbf{1}^TY) = Y^TH_1Y$$

$$\frac{\hat{\beta}^T X^T Y - n\bar{y}^2}{Y^T Y - n\bar{y}^2} = \frac{Y^T(H_x - H_1)Y}{Y^T(I - H_1)Y} = \frac{SSR}{SST} = R^2$$

$\blacksquare$

(c) $$\cos\theta = \frac{Y^T \hat{Y}}{\|Y\| \|\hat{Y}\|} = \frac{Y^T H_x Y}{\sqrt{Y^T Y}\sqrt{Y^T H_x Y}}$$

$$= \sqrt{\frac{Y^T H_x Y}{Y^T Y}} = R$$

$\blacksquare$

$\boxed{4}$.

(a) $\mathbb{E}Y = \mathbb{E}[\mathbb{E}[y|\gamma]] = \mathbb{E}[X\beta + Z\gamma] = X\beta$

$\mathbb{E}[\gamma] = 0$

$$\begin{aligned} \mathrm{var}(Y) &= \mathbb{E}[\mathrm{var}(Y|\gamma)] + \mathrm{var}(\mathbb{E}[Y|\gamma]) \\ &= \mathbb{E}[R] + \mathrm{var}(X\beta + Z\gamma) \\ &= R + ZDZ^T \end{aligned}$$

$\mathrm{var}(\gamma) = D$

$$\begin{aligned} \mathrm{cov}(y,\gamma) &= \mathbb{E}[y\gamma] - \mathbb{E}y\,\mathbb{E}\gamma \\ &= \mathbb{E}[\mathbb{E}[y\gamma|\gamma]] - 0 \\ &= \mathbb{E}[\gamma\,\mathbb{E}[y|\gamma]] \\ &= \mathbb{E}[X\beta\gamma + Z\gamma^2] \\ &= ZD \end{aligned}$$

$$\begin{pmatrix} y \\ \gamma \end{pmatrix} \sim N\left(\begin{pmatrix} X\beta \\ 0 \end{pmatrix}, \begin{pmatrix} ZDZ^T + R & ZD \\ DZ^T & D \end{pmatrix} \right)$$

(b) $X^TR^{-1}X\hat{\beta} + X^TR^{-1}Z\hat{\gamma} = X^TR^{-1}y$

$Z^TR^{-1}X\hat{\beta} + (D^{-1}+Z^TR^{-1}Z)\hat{\gamma} = Z^TR^{-1}y$

$$\Rightarrow \hat{\gamma} = \underbrace{(D^{-1}+Z^TR^{-1}Z)}_{=C}{}^{-1}Z^TR^{-1}(y-X\hat{\beta})$$

$X^TR^{-1}X\hat{\beta} + X^TR^{-1}Z\cdot CZ^TR^{-1}(y-X\hat{\beta}) = X^TR^{-1}y$

$$\underbrace{(X^TR^{-1}X - X^TR^{-1}ZCZ^TR^{-1}X)\hat{\beta}}_{= X^TR^{-1}\underbrace{(I-ZCZ^TR^{-1})X\hat{\beta}}_{=V^{-1}}} = X^T\underbrace{R^{-1}(I-ZCZ^TR^{-1})}_{=V^{-1}}y$$

$$\Rightarrow \hat{\beta} = (X^TV^{-1}X)^{-1}X^TV^{-1}y$$

$$\hat{\gamma} = (D^{-1}+Z^TR^{-1}Z)^{-1}Z^TR^{-1}(y-X\hat{\beta})$$

$$= DZ^T(ZDZ^T+R)^{-1}(y-X\hat{\beta})$$

$$= DZ^TV^{-1}(y-X\hat{\beta})$$

$\boxtimes$

$\boxed{5}$.

(a) $Q(\theta|\theta^{(p)}) = l(\theta) + \int \log k(Y_m|Y_o,\theta) k(Y_m|Y_o,\theta^{(p)})\, dY_m$

$\log f(Y_c|\theta^{(p+1)}) = \log g(Y_o|\theta^{(p+1)}) + \log k(Y_m|Y_o,\theta^{(p+1)})$

$$\Rightarrow \int \log f(Y_c|\theta^{(p+1)}) k(Y_m|Y_o,\theta^{(p)})\, dY_m$$
$$= \int \log g(Y_o|\theta^{(p+1)}) k(Y_m|Y_o,\theta^{(p)})\, dY_m$$
$$+ \int \log k(Y_m|Y_o,\theta^{(p+1)}) k(Y_m|Y_o,\theta^{(p)})\, dY_m$$

$$\int \log f(Y_c|\theta^{(p)}) k(Y_m|Y_o,\theta^{(p)})\, dY_m$$
$$= \int \log g(Y_o|\theta^{(p)}) k(Y_m|Y_o,\theta^{(p)})\, dY_m$$
$$+ \int \log k(Y_m|Y_o,\theta^{(p)}) k(Y_m|Y_o,\theta^{(p)})\, dY_m$$

$$\Rightarrow Q(\theta^{(p+1)}|\theta^{(p)}) = l(\theta^{(p+1)}) + H(\theta^{(p+1)}|\theta^{(p)})$$
$$Q(\theta^{(p)}|\theta^{(p)}) = l(\theta^{(p)}) + H(\theta^{(p)}|\theta^{(p)})$$

$$Q(\theta^{(p+1)}|\theta^{(p)}) - Q(\theta^{(p)}|\theta^{(p)})$$
$$= \left(l(\theta^{(p+1)}) - l(\theta^{(p)})\right) + \underbrace{\left(H(\theta^{(p+1)}|\theta^{(p)}) - H(\theta^{(p)}|\theta^{(p)})\right)}_{\leq 0} \geq 0$$

$$\int \log \frac{k(Y_m|Y_o,\theta^{(p+1)})}{k(Y_m|Y_o,\theta^{(p)})} \times k(Y_m|Y_o,\theta^{(p)})\, dY_m$$
$$\underset{\text{J.Ineq}}{\leq} \log \int k(Y_m|Y_o,\theta^{(p+1)})\, dY_m = 0$$

$$\Rightarrow l(\theta^{(p+1)}) \geq l(\theta^{(p)})$$

$\blacksquare$

(b) $f(y_i|\theta) = \sum_{k=1}^{K} p_k \frac{1}{\sigma_k} \phi\left(\frac{y_i-\mu_k}{\sigma_k}\right)$

$$Z_{ik} = \begin{cases} 1 & , i=k \\ 0 & , o/w \end{cases}$$

$\sum_{k=1}^{K} Z_{ik} = 1 \Rightarrow \mathbb{P}(Z_{ik}=1) = p_k$

$Y_i | Z_{ik}=1 \sim N(\mu_k, \sigma_k^2)$

complete data $(Y_i, Z_{i1}, \dots Z_{iK})$, $i=1\dots n$.

$$L_c(\theta) = \prod_{i=1}^{n} \prod_{k=1}^{K} \left[p_k \frac{1}{\sigma_k} \phi\left(\frac{y_i-\mu_k}{\sigma_k}\right)\right]^{Z_{ik}}$$

$$l_c(\theta) = \sum_{i=1}^{n} \sum_{k=1}^{K} Z_{ik}\left[\log p_k - \log \sigma_k + \log \phi\left(\frac{y_i-\mu_k}{\sigma_k}\right)\right]$$

상수항 제외 $\rightarrow$ $$= \sum_{i=1}^{n} \sum_{k=1}^{K} Z_{ik}\left[\log p_k - \frac{1}{2}\log \sigma_k^2 - \frac{(y_i-\mu_k)^2}{2\sigma_k^2}\right]$$

E-step

$$\theta^{(p)} = (\mu_1^{(p)}, \dots, \mu_K^{(p)}, \sigma_1^{2(p)}, \dots, \sigma_K^{2(p)}, p_1^{(p)}, \dots, p_K^{(p)})$$

$$w_{ik}^{(p)} := \mathbb{E}[Z_{ik} | y_i, \theta^{(p)}]$$

$$= \mathbb{P}(Z_{ik}=1 | y_i, \theta^{(p)})$$

$$= \frac{p_k^{(p)} \frac{1}{\sigma_k^{(p)}} \phi\left(\frac{y_i-\mu_k^{(p)}}{\sigma_k^{(p)}}\right)}{\sum_{l=1}^{K} p_l^{(p)} \frac{1}{\sigma_l^{(p)}} \phi\left(\frac{y_i-\mu_l^{(p)}}{\sigma_l^{(p)}}\right)}$$

$$Q(\theta|\theta^{(p)}) = \mathbb{E}[l_c(\theta) | y_1, \dots y_n, \theta^{(p)}]$$

$$= \sum_{i=1}^{n} \sum_{k=1}^{K} w_{ik}^{(p)}\left[\log p_k - \frac{1}{2}\log \sigma_k^2 - \frac{(y_i-\mu_k)^2}{2\sigma_k^2}\right]$$

$\uparrow$
상수항 생략

M-Step

$$\sum_{k=1}^{K} p_k = 1 \Rightarrow \sum_{i=1}^{n}\sum_{k=1}^{K} w_{ik}^{(p)} \log p_k$$

using Lagrange multiplier,

$$\sum_{i=1}^{n} \frac{w_{ik}^{(p)}}{p_k} - \lambda = 0 \Rightarrow p_k = \frac{1}{\lambda}\sum_{i=1}^{n} w_{ik}^{(p)}$$

$$\Rightarrow 1 = \sum_{k=1}^{K} p_k = \frac{1}{\lambda}\sum_{k=1}^{K}\sum_{i=1}^{n} w_{ik}^{(p)} \Rightarrow \lambda = n$$

$$\sum_{k=1}^{K} w_{ik}^{(p)} = 1$$

$$p_k^{(p+1)} = \frac{1}{n}\sum_{i=1}^{n} w_{ik}^{(p)}$$

- update μ_k

$$-\frac{1}{2}\sum_{i=1}^{n} w_{ik}^{(p)} \frac{(y_i-\mu_k)^2}{\sigma_k^2}$$

$$\xrightarrow{\frac{\partial}{\partial \mu_k}} \sum_{i=1}^{n} w_{ik}^{(p)} (y_i - \mu_k) = 0$$

$$\Rightarrow \mu_k^{(p+1)} = \frac{\sum_{i=1}^{n} w_{ik}^{(p)} y_i}{\sum_{i=1}^{n} w_{ik}^{(p)}}$$

- update σ_k^2

$$-\frac{1}{2}\sum_{i=1}^{n} w_{ik}^{(p)} \left[\log \sigma_k^2 + \frac{(y_i-\mu_k)^2}{\sigma_k^2}\right]$$

$$\xrightarrow{\frac{\partial}{\partial \sigma_k^2}} -\frac{1}{2}\frac{\sum w_{ik}^{(p)}}{\sigma_k^2} + \frac{1}{2}\frac{\sum_{i=1}^{n} w_{ik}^{(p)} (y_i-\mu_k)^2}{(\sigma_k^2)^2} = 0$$

$$\Rightarrow \sigma_k^2 = \frac{\sum_{i=1}^{n} w_{ik}^{(p)} (y_i-\mu_k)^2}{\sum_{i=1}^{n} w_{ik}^{(p)}} = \frac{\sum_{i=1}^{n} w_{ik}^{(p)} (y_i-\mu_k^{(p+1)})^2}{\sum_{i=1}^{n} w_{ik}^{(p)}}$$